

Socioeconomic Status and Psychological Distress Among Primary Caregivers in Rural China

A cross sectional analysis of the 2023 Ya'an baseline survey

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Abstract

This study examines whether household income and caregiver education are associated with self reported psychological distress among primary caregivers of young children in rural Ya'an, Sichuan, China. It uses an archived 2023 baseline survey extract containing 919 caregivers. Separate ordinary least squares models relate stored logarithmic DASS outcomes to household income and caregiver education, with adjustment for household amenities, caregiver age and health, and selected child characteristics. The analysis is observational; the wider parenting intervention project does not randomize socioeconomic status.

In the full extract, the adjusted income coefficient is -0.06353 (HC1 standard error 0.01721), and the education coefficient is -0.13209 (0.04000). Negative coefficients also appear across the stress, anxiety and depression outcomes. However, the income estimate is sensitive to 16 records whose stored log income equals zero: excluding these records yields -0.02334 (0.04651). Their original coding remains unresolved, so this restriction is an audit diagnostic rather than a preferred sample. Education estimates remain negative in the selected coding checks. Subgroup estimates are exploratory and do not establish differences between caregiver groups.

The results suggest a socioeconomic pattern in reported distress within this survey extract, with stronger consistency for education than for income across the checks performed. They do not establish that increasing income or education would improve mental health. Interpretation also depends on unresolved questionnaire scoring and sample construction, and reported standard errors cannot account for village clustering. The study identifies priorities for a documented reanalysis and for research on the resources and conditions surrounding caregiving.

Keywords: caregiver mental health; socioeconomic status; rural China; education; household income; psychological distress

1 Introduction

Caregiving takes place within a household and a wider community. Income can affect the resources a family can draw on, while education may be associated with access to information, work opportunities and ways of navigating services. These circumstances make the relationship between socioeconomic status and caregiver mental health an important question. They also make it difficult to interpret a single observed association: family resources, health and caregiving responsibilities may influence one another.

This thesis focuses on primary caregivers of young children in rural Ya'an. It asks whether higher household income and caregiver education are associated with lower reported depression, anxiety and stress symptoms in the 2023 baseline survey. The question is motivated by the socioeconomic gradient in health described by Adler et al. (1994), and by research on caregiving and early childhood development. The empirical analysis concerns the caregivers themselves. It does not estimate effects on their children or evaluate the effectiveness of parenting centers.

The study contributes a local descriptive analysis rather than a causal test. Its value lies in examining how socioeconomic indicators and distress vary together, making the measurement choices visible, and identifying where the available evidence is fragile. This distinction matters for policy. An association may help frame further investigation, but it does not by itself show which intervention would work or which families should be targeted.

The revised manuscript retains the original sequence of literature review, sample description, model specification, descriptive analysis, regressions, subgroup analysis and sensitivity checks. It replaces inconsistent model notation and unsupported interpretations, and recomputes the tables from the available 919 record analytical worksheet. The original income coefficients and sample counts can be reproduced to the precision printed in the 2024 thesis. The original education models used 922 records; that larger extract has not been reconstructed, so the revised education tables explicitly use the common 919 record sample.

2 Literature review

2.1 Socioeconomic circumstances and caregiver mental health

Adler et al. (1994) describe a graded association between socioeconomic status and health across the social hierarchy. Their discussion also emphasizes the difficulty of identifying the processes behind that relationship. Income and education therefore provide useful indicators of circumstances, but neither should be treated as a complete measure of socioeconomic position. Household income may vary over time; educational attainment may capture experiences that preceded current caregiving responsibilities.

Gallo and Matthews (1999) examine whether negative emotions contribute to the relationship between socioeconomic status and health. They discuss evidence linking socioeconomic circumstances, emotional experiences and health, while stressing the limits of indirect evidence about mediation. For the present study, financial strain and access to supportive resources are

possible explanations for an association. They are not mechanisms identified by the regressions, because the extract does not contain a documented set of measures that would test those pathways.

Caregiver research also shows why the population and caregiving context matter. Bin Sallim et al. (2015) synthesize evidence on mental health disorders among caregivers of people with Alzheimer disease. Oshio (2014) uses a Japanese panel to examine family caregiving and psychological distress. These studies concern different ages, care recipients and social settings from the Ya'an sample. They motivate attention to caregiving demands, but their findings cannot be transferred directly to caregivers of young children in rural China. In particular, the longitudinal fixed effects approach used by Oshio is not the design used in this thesis.

Subjective social status offers a further perspective. Demakakos et al. (2008) examine perceived social standing alongside objective socioeconomic indicators in older adults in England. Their results suggest that material resources and perceived position may capture different aspects of a person's circumstances. The present extract does not include a verified subjective status measure. It therefore cannot determine whether income or education is operating through perceived status, social comparison or another process. Nor does it establish a general contrast between Chinese and Western cultural responses.

2.2 Early childhood development and rural China

The early childhood literature places caregiving within the conditions that support children's development. Black et al. (2017) discuss the importance of early environments and the uneven reach of developmental programs. This literature helps explain why caregiver well being is relevant to family support, but relevance is not the same as evidence that the socioeconomic associations estimated here affect child development.

Emmers et al. (2021) review early development and parental training in rural China. Across the included study sites, they report substantial risks of developmental delay and evidence that parental training can improve developmental outcomes. These are estimates from the studies in their review, not prevalence estimates for the Ya'an caregivers or proof that a particular local parenting center improved mental health. The distinction between the observational baseline used here and an intervention evaluation is essential.

Zhang et al. (2018) study caregiver symptoms and early childhood development across several rural communities in China. Their findings motivate attention to both caregiver well being and developmental environments. They do not imply that every caregiver meeting a symptom threshold has a clinical disorder, or that an observed caregiver-child association identifies a causal pathway. This thesis uses caregiver DASS outcomes only and does not reproduce their child development analysis.

Taken together, the literature provides a reason to examine caregiver circumstances while keeping several questions separate: whether symptoms vary with socioeconomic indicators, what processes explain that variation, and whether changing a resource or service improves outcomes. The present

cross sectional analysis addresses the first question. Longitudinal measurement and suitable intervention designs would be needed to address the others.

3 Sample and study context

The original thesis describes a baseline survey collected in 2023 in rural Ya'an, Sichuan, within a wider early childhood and parenting intervention project. Its study description refers to 12 villages across five counties, families with children aged approximately 6 to 36 months, and an initial baseline of about 1,100 caregivers. These details are retained as the original study context. The current analytical extract does not independently verify recruitment probabilities, the complete recruitment roster or the sequence of exclusions.

Participation in a wider randomized project does not make the income or education comparisons random. This analysis uses baseline differences between caregivers and households. No treatment assignment, follow up outcome or intervention effect is estimated. Selection of villages and eligible families may also limit generalization. The fact that a region is predominantly Han or includes economically disadvantaged communities does not establish that it represents rural China as a whole.

For this revision, the source is the analytical data worksheet in the archived file Empirical data (1).xlsx. It contains 919 distinct analytical records and no missing values in the outcomes, focal variables or controls used below. There are 89 fathers, 654 mothers and 176 grandparents according to the archived caregiver role coding. These are analytical sample counts, not an attrition flow from the initial survey.

The extract reproduces the original income regressions, including the component, caregiver subgroup and recorded age restriction models, to printed rounding. The original education tables report 922 observations and slightly different coefficients. Rather than attach that unreconstructed sample count to the available data, all revised main tables use the same 919 record extract. The manuscript makes no assumption that the additional three records were invalid or that the common sample is more representative.

4 Measures and model specification

4 1 Outcomes and variable construction

The DASS measures self reported depression, anxiety and stress symptoms rather than establishing a clinical diagnosis (Lovibond and Lovibond, 1995; University of New South Wales, n.d.). The source thesis identifies the questionnaire as DASS 21. This revision analyzes the four stored transformed outcomes named `dass_total`, `dass_stress`, `dass_anxiety` and `dass_depression`. Higher values are interpreted as greater reported symptom burden within the archived coding, subject to the measurement audit described below. They are not measures of happiness.

The worksheet contains transformed outcomes rather than the questionnaire items. Its values are consistent with logarithmic score transformations, but the exact item summation, multiplier, offset

and sequence of truncation cannot be established from this file alone. For 905 records, exponentiating the stored total agrees, within an absolute tolerance of 0.00001, with the sum of the exponentiated subscales minus two. This is compatible with an offset of one applied before logarithms. Fourteen records do not satisfy that identity. This calculation is a diagnostic, not confirmation of the raw scoring rule; Appendix A documents the unresolved construction issues.

For that reason, the descriptive table reports the stored transformed variables directly. It does not back transform them into certified raw DASS scores, apply clinical cutoffs or estimate disorder prevalence. The original raw income standard deviation was internally inconsistent with its stated range. That statistic is removed rather than replaced with a guessed decimal. Income and age are instead described on their stored logarithmic scales, which are the variables used in the regressions.

The focal predictors are the archived household income variable income, described as logarithmic income in the original documentation, and caregiver education edu. Education takes codes from zero to five; the original legend describes codes one to five as successive schooling categories, from no schooling through college or above. Code zero is not defined in the recovered legend. The linear education coefficient therefore represents a one code step, not one year of schooling. It imposes equal spacing between categories, an assumption that should be tested once the questionnaire coding is recovered.

Sixteen income values equal zero on the stored log scale, and 23 education values use code zero. A log income value of zero could reflect a low positive amount, an offset or a special coding convention; it must not automatically be called a missing value or a genuinely zero income household. The primary analysis retains all records as stored. Additional exclusions are clearly labeled coding diagnostics and are not chosen to strengthen the findings.

The adjustment variables follow the original analysis: internet access, air conditioning, a village doctor indicator, car ownership, stored log caregiver age, self reported health, a child sex indicator and an ordinal code for expected child education. The child sex variable is not caregiver gender. These variables describe selected household and caregiver characteristics; they do not exhaust possible confounders. Appendix A gives the recovered definitions and coding limits.

4 2 Cross sectional ordinary least squares models

Separate regressions are estimated for household income and caregiver education. For caregiver i , let Y be a stored transformed distress outcome, L the stored log income variable, E the education code, and X the vector of adjustment variables. The baseline specifications are

$$Y_i = \alpha + \beta L_i + X_i' \gamma + \varepsilon_i \quad (1)$$

$$Y_i = \delta + \theta E_i + X_i' \kappa + u_i \quad (2)$$

The specifications are repeated for the stress, anxiety and depression outcomes. Unadjusted models include only the focal predictor and an intercept. There is one cross sectional baseline record per analytical caregiver, so the model does not include individual or time fixed effects. Separate income

and education models also do not hold the other socioeconomic indicator constant; a joint model is reported as an additional specification check.

All revised regression tables use ordinary least squares and HC1 heteroskedasticity robust standard errors. Tests and confidence intervals use the residual degrees of freedom. The original unadjusted income coefficient is unchanged, but its uncertainty is recomputed under this common robust convention rather than retaining the original conventional standard error. Parentheses now consistently contain nonnegative standard errors, not the test statistics that were mislabeled in parts of the original manuscript.

Caregivers recruited within villages may share unobserved conditions. The recovered worksheet has no usable village identifier, so these standard errors cannot account for within village dependence. With the original study description referring to only 12 villages, a further reanalysis would also need an inference approach suited to a small number of clusters. Reported p values are consequently conditional on the current variance specification. They should not be treated as definitive evidence after village clustering or multiple testing.

The coefficients describe associations conditional on the included controls. Reverse causation, omitted family circumstances, measurement error and selection remain possible. For example, health or family expectations may be related to both distress and socioeconomic circumstances. Adjustment does not resolve their direction of influence. Without a verified raw transformation, the income coefficient is also not presented as a percentage change in clinical symptoms or a causal elasticity.

5 Descriptive analysis

Table 1 summarizes the variables actually used in the revised regressions. All rows refer to the same 919 records. The mean stored total outcome is 2.401 and its median is 2.708; the component distributions differ. These summaries describe the analytical variables and do not determine how many caregivers have a disorder. Means and medians alone are insufficient to classify either clinical severity or the full shape of a distribution.

Table 1 Descriptive statistics for the revised analytical sample

Variable	N	Mean	Median	SD	Min	Max
Total distress (stored log)	919	2.401	2.708	1.208	0.000	4.419
Stress (stored log)	919	1.780	1.946	1.045	0.000	3.761
Anxiety (stored log)	919	1.329	1.609	0.997	0.000	3.761
Depression (stored log)	919	1.291	1.099	1.059	0.000	3.761
Household income (stored log)	919	10.429	10.779	1.624	0.000	12.429

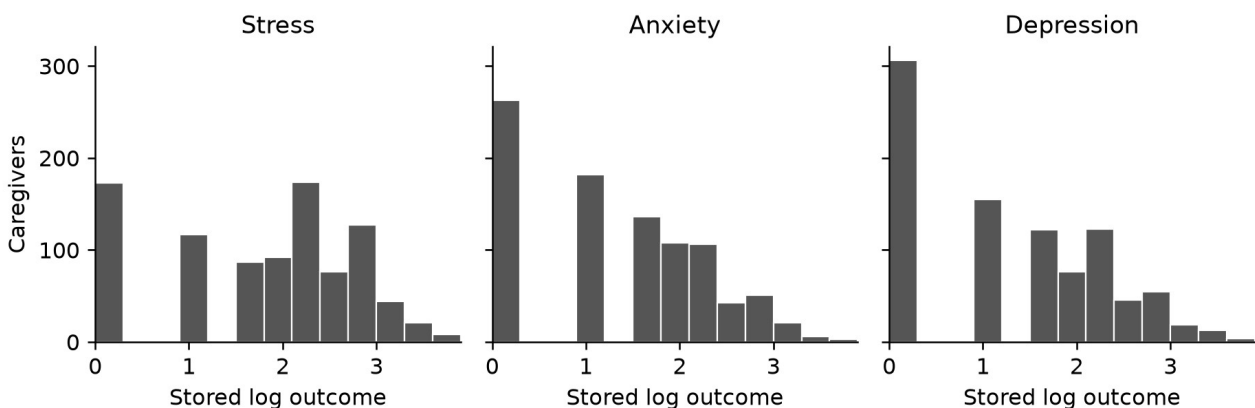
Variable	N	Mean	Median	SD	Min	Max
Caregiver education (ordinal code)	919	2.435	2.000	1.082	0.000	5.000
Internet access	919	0.974	1.000	0.160	0.000	1.000
Air conditioning	919	0.621	1.000	0.485	0.000	1.000
Village doctor indicator	919	0.303	0.000	0.460	0.000	1.000
Household car ownership	919	0.756	1.000	0.430	0.000	1.000
Caregiver age (stored log)	919	3.474	3.401	0.287	2.944	4.094
Self reported health indicator	919	0.638	1.000	0.481	0.000	1.000
Child sex indicator	919	0.535	1.000	0.499	0.000	1.000
Child education expectations (ordinal code)	919	2.449	3.000	0.704	1.000	3.000

Source: Archived 919 record analytical worksheet. Natural log labels follow the original variable documentation; exact raw score offsets and preprocessing remain unverified. Binary and ordinal codes are summarized numerically without treating them as continuous physical quantities.

The mean stored log income is 10.429 and the median is 10.779. A small group of observations lies at the lower bound of zero, which motivates the income coding check below. Education is concentrated in the middle categories: 400 records have code two and 213 have code three. Code zero occurs in 23 records. Internet access is common in this extract, while the village doctor indicator varies across households. These are sample descriptions, not estimates of regional coverage.

Figure 1 shows the stored log stress, anxiety and depression distributions using common axes. These transformed variables summarize symptom patterns in the extract; they cannot be used here to estimate the prevalence of diagnosed conditions.

Figure 1 Distributions of stored component outcomes



Source: Archived analytical worksheet, N = 919. The horizontal axes display transformed values as stored; all panels use common bins and a common vertical scale.

5 1 Collinearity and correlations

Table 2 recomputes variance inflation factors for the predictors in each adjusted model. This expands the original diagnostic, which listed controls but omitted the focal socioeconomic predictor. The values are modest within these particular designs. That finding does not establish correct measurement, eliminate omitted variable bias or justify a causal interpretation.

Table 2 Variance inflation factors in the adjusted models

Predictor	Income model	Education model
Household income (stored log)	1.09	—
Caregiver education (ordinal code)	—	1.31
Internet access	1.06	1.06
Air conditioning	1.07	1.07
Village doctor indicator	1.02	1.02
Household car ownership	1.11	1.11
Caregiver age (stored log)	1.17	1.36
Self reported health indicator	1.11	1.11
Child sex indicator	1.01	1.01
Child education expectations (ordinal code)	1.04	1.03

An intercept is included when calculating each VIF. A dash denotes a predictor absent from that separate model. These diagnostics concern linear dependence among observed predictors only.

Table 3 presents selected pairwise correlations from the common sample. Income and education are each negatively correlated with the stored total outcome and positively correlated with one another. This pattern motivates the regressions, but a correlation cannot identify the direction of a relationship. The table does not serve as a test of whether the survey data are valid. Stars use the same significance order throughout the manuscript.

Table 3 Selected Pearson correlations

Variable	Total distress (stored log)	Income (stored log)	Education (code)
Household income (stored log)	-0.097***	1.000	0.163***
Caregiver education (ordinal code)	-0.099***	0.163***	1.000
Internet access	-0.070**	0.040	0.060*

Variable	Total distress (stored log)	Income (stored log)	Education (code)
Air conditioning	-0.064*	0.121***	0.105***
Village doctor indicator	-0.022	0.018	0.031
Household car ownership	-0.012	0.156***	0.191***
Caregiver age (stored log)	-0.036	-0.201***	-0.451***
Self reported health indicator	-0.220***	0.165***	0.222***
Child sex indicator	0.049	0.004	-0.020
Child education expectations (ordinal code)	-0.060*	-0.117***	-0.101***

N = 919 for every pair. * $p < .10$; ** $p < .05$; *** $p < .01$. Correlation tests assume independent records and are exploratory. Binary and ordinal variables are shown using their stored numerical codes.

6 Empirical analysis

6.1 Household income and total distress

Table 4 reports unadjusted and adjusted associations between stored log household income and the total distress outcome. The adjusted coefficient is -0.06353 , with a robust standard error of 0.01721 and a 95 percent confidence interval from -0.09730 to -0.02976 . Both the coefficient and the 919 observation count reproduce the original adjusted income table. Adding controls changes the coefficient modestly and raises R squared from 0.009 to 0.076 .

Table 4 Household income and total distress

Variable	Unadjusted	Adjusted
Household income (stored log)	-0.07232*** (0.01755)	-0.06353*** (0.01721)
Internet access	—	-0.46736** (0.20125)
Air conditioning	—	-0.09969 (0.08148)
Village doctor indicator	—	-0.01333 (0.08590)
Household car ownership	—	0.05236 (0.09571)
Caregiver age (stored log)	—	-0.48640*** (0.14862)
Self reported health indicator	—	-0.57411*** (0.08276)

Variable	Unadjusted	Adjusted
Child sex indicator	—	0.14431* (0.07738)
Child education expectations (ordinal code)	—	-0.10200* (0.05624)
Constant	3.15534*** (0.18248)	5.77348*** (0.63562)
Observations	919	919
R squared	0.009	0.076

Notes: Coefficients with HC1 heteroskedasticity robust standard errors in parentheses. Two sided t tests use residual degrees of freedom. * $p < .10$; ** $p < .05$; *** $p < .01$. Standard errors do not account for village clustering. All estimates are recomputed from the stored analytical worksheet; subgroup and multiple outcome tests are exploratory.

The negative coefficient indicates that higher values of the stored income variable accompany lower values of the stored distress outcome within this sample and adjustment set. It does not show that income improves mental health independently of all other circumstances. The separate income model does not adjust for caregiver education, and the coding check in Section 7 shows that the income result is sensitive to the records at zero on the log scale.

The adjusted coefficients for stored log age and the health indicator are also negative. These results concern conditional associations in the pooled sample, not the effect of aging or proof that older caregivers are happier. The model explains a limited share of variation. A low R squared does not by itself invalidate an association, but it makes clear that these observed variables leave much of the variation unexplained.

6 2 Household income and component outcomes

Table 5 repeats the income specification for stress, anxiety and depression. The income coefficients are -0.03978 , -0.05098 and -0.05468 , respectively. These coefficients and sample counts also reproduce the original printed income component results. All three estimated associations are negative under the reported variance convention, but the table contains multiple related exploratory tests rather than three independent confirmations of a causal effect.

Table 5 Household income and component outcomes

Variable	Stress	Anxiety	Depression
Household income (stored log)	-0.03978** (0.01585)	-0.05098*** (0.01579)	-0.05468*** (0.02055)
Internet access	-0.28693 (0.19494)	-0.37045** (0.18775)	-0.50600*** (0.19372)
Air conditioning	-0.08751 (0.07024)	-0.13322** (0.06788)	-0.06740 (0.07178)

Variable	Stress	Anxiety	Depression
Village doctor indicator	0.02896 (0.07214)	-0.00682 (0.07064)	-0.02152 (0.07575)
Household car ownership	0.03897 (0.08486)	0.05163 (0.08043)	0.00370 (0.08355)
Caregiver age (stored log)	-0.37648*** (0.13215)	-0.30344** (0.12107)	-0.47842*** (0.12824)
Self reported health indicator	-0.44171*** (0.07166)	-0.46868*** (0.06960)	-0.46433*** (0.07360)
Child sex indicator	0.10269 (0.06722)	0.06758 (0.06389)	0.07772 (0.06783)
Child education expectations (ordinal code)	-0.11080** (0.04924)	-0.06865 (0.04685)	-0.09153* (0.04966)
Constant	4.29664*** (0.57915)	3.75206*** (0.52162)	4.53969*** (0.58204)
Observations	919	919	919
R squared	0.059	0.073	0.070

Notes: Coefficients with HC1 heteroskedasticity robust standard errors in parentheses. Two sided t tests use residual degrees of freedom. * $p < .10$; ** $p < .05$; *** $p < .01$. Standard errors do not account for village clustering. All estimates are recomputed from the stored analytical worksheet; subgroup and multiple outcome tests are exploratory.

Financial strain, the availability of services and access to supportive activities could help explain a socioeconomic pattern in distress. The available regressions do not distinguish these explanations. Nor do the coefficient magnitudes establish which symptom domain is most responsive to income, because outcome construction and dependence between the measures would need further examination.

6 3 Caregiver education and total distress

Table 6 uses the same 919 records to examine education. The adjusted education coefficient is -0.13209 , with a robust standard error of 0.04000 and a 95 percent confidence interval from -0.21059 to -0.05359 . The original 922 observation model reported -0.12746 . These are estimates from different analytical extracts; the revised number is a recomputation, not a correction to the last decimal of the original estimate.

Table 6 Caregiver education and total distress

Variable	Unadjusted	Adjusted
Caregiver education (ordinal code)	-0.11100*** (0.03644)	-0.13209*** (0.04000)
Internet access	—	-0.48328** (0.20415)

Variable	Unadjusted	Adjusted
Air conditioning	—	-0.09696 (0.08090)
Village doctor indicator	—	-0.01826 (0.08518)
Household car ownership	—	0.06388 (0.09518)
Caregiver age (stored log)	—	-0.63700*** (0.15949)
Self reported health indicator	—	-0.56767*** (0.08312)
Child sex indicator	—	0.14089* (0.07717)
Child education expectations (ordinal code)	—	-0.09647* (0.05590)
Constant	2.67138*** (0.09663)	5.94648*** (0.65843)
Observations	919	919
R squared	0.010	0.080

Notes: Coefficients with HC1 heteroskedasticity robust standard errors in parentheses. Two sided t tests use residual degrees of freedom. * $p < .10$; ** $p < .05$; *** $p < .01$. Standard errors do not account for village clustering. All estimates are recomputed from the stored analytical worksheet; subgroup and multiple outcome tests are exploratory.

Higher education codes are associated with lower caregiver distress, conditional on the included controls. The outcome belongs to the caregiver, not to the child or another person under their care. A one code increment is not equivalent to an additional year of education. The result also does not establish that an educational program delivered to caregivers would improve mental health. Prior circumstances that shaped educational attainment may remain relevant.

6 4 Caregiver education and component outcomes

Table 7 reports the component models for education in the common sample. Each estimated education coefficient is negative. This pattern is consistent with the total outcome result, but the equal spacing assumption for education codes and the unresolved meaning of code zero remain relevant. The components describe reported symptoms; they do not establish diagnoses or measure children's developmental outcomes.

Table 7 Caregiver education and component outcomes

Variable	Stress	Anxiety	Depression
Caregiver education (ordinal code)	-0.07783** (0.03542)	-0.12299*** (0.03460)	-0.07768** (0.03546)
Internet access	-0.29572 (0.19317)	-0.38730** (0.19335)	-0.51106*** (0.19422)
Air conditioning	-0.08655 (0.07011)	-0.12842* (0.06738)	-0.07057 (0.07172)
Village doctor indicator	0.02615 (0.07174)	-0.01173 (0.06997)	-0.02373 (0.07534)
Household car ownership	0.04476 (0.08479)	0.06582 (0.07921)	0.00314 (0.08298)
Caregiver age (stored log)	-0.46330*** (0.14207)	-0.45034*** (0.12949)	-0.55282*** (0.13806)
Self reported health indicator	-0.43882*** (0.07181)	-0.45953*** (0.06936)	-0.46721*** (0.07400)
Child sex indicator	0.10067 (0.06716)	0.06443 (0.06365)	0.07566 (0.06798)
Child education expectations (ordinal code)	-0.10705** (0.04899)	-0.06519 (0.04637)	-0.08469* (0.04937)
Constant	4.36733*** (0.59316)	4.02180*** (0.54582)	4.41126*** (0.58269)
Observations	919	919	919
R squared	0.061	0.081	0.068

Notes: Coefficients with HC1 heteroskedasticity robust standard errors in parentheses. Two sided t tests use residual degrees of freedom. * $p < .10$; ** $p < .05$; *** $p < .01$. Standard errors do not account for village clustering. All estimates are recomputed from the stored analytical worksheet; subgroup and multiple outcome tests are exploratory.

Education could be related to work opportunities, access to information or ways of finding help. Emotional intelligence, coping ability and social support are not measured well enough here to determine whether they mediate the association. They remain questions for further research rather than explanations demonstrated by Table 7.

6 5 Caregiver subgroup analysis

Tables 8 and 9 report exploratory models for fathers, mothers and grandparents. The groups are defined by the recorded caregiving relationship, rather than being inferred solely from gender or age. Their sample sizes differ substantially: 89 fathers, 654 mothers and 176 grandparents. All education subgroup models now use the same extract as the income models.

Table 8 Household income and total distress by caregiver role

Variable	Fathers	Mothers	Grandparents
Household income (stored log)	-0.04238 (0.04652)	-0.03970 (0.02722)	-0.08725*** (0.03012)
Internet access	-0.74387** (0.34020)	-0.40900 (0.27899)	-0.51148 (0.32731)
Air conditioning	-0.40297* (0.24037)	-0.03191 (0.09757)	-0.21335 (0.21160)
Village doctor indicator	-0.01587 (0.23214)	-0.02042 (0.09766)	0.01154 (0.29831)
Household car ownership	0.26915 (0.25222)	-0.03423 (0.11874)	0.19830 (0.22507)
Caregiver age (stored log)	0.19049 (0.65103)	-0.49437* (0.29319)	-0.01527 (1.39235)
Self reported health indicator	-0.55382** (0.25572)	-0.52365*** (0.09904)	-0.73511*** (0.20641)
Child sex indicator	0.23664 (0.25139)	0.15267* (0.09244)	0.14417 (0.19698)
Child education expectations (ordinal code)	-0.13577 (0.14183)	-0.10176 (0.06575)	-0.02707 (0.17023)
Constant	3.45831 (2.43788)	5.48863*** (1.04463)	4.01558 (5.83655)
Observations	89	654	176
R squared	0.151	0.058	0.134

Notes: Coefficients with HC1 heteroskedasticity robust standard errors in parentheses. Two sided t tests use residual degrees of freedom. * $p < .10$; ** $p < .05$; *** $p < .01$. Standard errors do not account for village clustering. All estimates are recomputed from the stored analytical worksheet; subgroup and multiple outcome tests are exploratory.

Table 9 Education and total distress by caregiver role

Variable	Fathers	Mothers	Grandparents
Caregiver education (ordinal code)	0.15915 (0.13281)	-0.14039*** (0.04583)	-0.25818** (0.11271)
Internet access	-0.47064 (0.45792)	-0.45448 (0.27883)	-0.40744 (0.34554)
Air conditioning	-0.47579** (0.22814)	-0.01686 (0.09681)	-0.18334 (0.20985)
Village doctor indicator	-0.01274 (0.22667)	-0.03321 (0.09684)	0.04160 (0.29342)
Household car ownership	0.21215 (0.25998)	-0.00255 (0.11744)	0.19401 (0.22190)

Variable	Fathers	Mothers	Grandparents
Caregiver age (stored log)	0.46598 (0.65988)	-0.58503** (0.29200)	0.44504 (1.33767)
Self reported health indicator	-0.59897** (0.25882)	-0.50649*** (0.09791)	-0.68597*** (0.20970)
Child sex indicator	0.30183 (0.24259)	0.15263* (0.09177)	0.14621 (0.20105)
Child education expectations (ordinal code)	-0.15945 (0.14048)	-0.10802* (0.06476)	-0.00361 (0.16910)
Constant	1.55232 (2.50666)	5.76817*** (1.05411)	1.50653 (5.46878)
Observations	89	654	176
R squared	0.157	0.070	0.134

Notes: Coefficients with HC1 heteroskedasticity robust standard errors in parentheses. Two sided t tests use residual degrees of freedom. * $p < .10$; ** $p < .05$; *** $p < .01$. Standard errors do not account for village clustering. All estimates are recomputed from the stored analytical worksheet; subgroup and multiple outcome tests are exploratory.

The estimated income coefficient is most negative in the grandparent group. Education coefficients are negative for mothers and grandparents, while the father estimate is less precise. These differences in point estimates and significance do not establish that the underlying associations differ across groups. A pooled model with interactions and a formal test would be required for that comparison; small subgroup samples and village dependence would still need attention.

The intercepts must not be used to rank the groups' mental health. They describe fitted outcomes when the numerical predictors are zero, including a stored log age of zero, a reference point outside the observed caregiver ages. They are not adjusted group means. Likewise, a nonsignificant father coefficient is not evidence that education or income is irrelevant to fathers. The data may be too imprecise to separate an association from zero in that subgroup.

7 Sensitivity analysis and coding audit

7 1 Recorded age restriction

The original analysis used a flag named tc to exclude some caregivers below sex specific marriage age thresholds. The recovered notes describe thresholds of 22 for men and 20 for women, but the exact creation of that flag is not independently reconstructed here. Table 10 therefore applies the recorded rule $tc = 0$ without claiming a newly verified legal classification. It leaves 907 records in the common sample for both revised models.

Table 10 Models restricted by the archived age flag

Variable	Income model	Education model
Household income (stored log)	-0.06336*** (0.01722)	—
Caregiver education (ordinal code)	—	-0.13113*** (0.04073)
Internet access	-0.46541** (0.20164)	-0.48109** (0.20468)
Air conditioning	-0.09949 (0.08264)	-0.10008 (0.08205)
Village doctor indicator	-0.01714 (0.08700)	-0.02166 (0.08628)
Household car ownership	0.05085 (0.09668)	0.06322 (0.09625)
Caregiver age (stored log)	-0.48398*** (0.15431)	-0.64677*** (0.16807)
Self reported health indicator	-0.57199*** (0.08387)	-0.56729*** (0.08416)
Child sex indicator	0.14029* (0.07818)	0.13666* (0.07798)
Child education expectations (ordinal code)	-0.09756* (0.05702)	-0.09069 (0.05665)
Constant	5.75313*** (0.65571)	5.96889*** (0.69210)
Observations	907	907
R squared	0.074	0.078

Notes: Coefficients with HC1 heteroskedasticity robust standard errors in parentheses. Two sided t tests use residual degrees of freedom. * $p < .10$; ** $p < .05$; *** $p < .01$. Standard errors do not account for village clustering. All estimates are recomputed from the stored analytical worksheet; subgroup and multiple outcome tests are exploratory.

The income coefficient remains close to the full sample result, and the education association remains negative. This is a sample sensitivity check. It does not show that younger caregivers lack caregiving competence or that excluding them makes the model valid. It also cannot resolve the income coding issue, omitted factors or clustering. The restricted sample is not preferred merely because its coefficient retains a significance star.

7 2 Income and education coding checks

Table 11 examines unresolved features of the archived variables. The full sample estimates are shown alongside models that omit the 16 records at zero on the stored log income scale, omit the 23 records with an undefined education code, or include income and education together. These checks

were added during the 2026 audit after inspecting the data. They are not preregistered robustness tests, and their p values are not adjusted for the exploratory set of comparisons.

Table 11 Additional specification and coding diagnostics

Check	N	Income coefficient (HC1 SE)	Education coefficient (HC1 SE)
Full archived sample	919	-0.06353 (0.01721)	-0.13209 (0.04000)
Omit stored log income = 0	903	-0.02334 (0.04651)	-0.13605 (0.04031)
Omit undefined education code 0	896	-0.06199 (0.01737)	-0.12691 (0.04329)
Include both SES predictors	919	-0.06022 (0.01746)	-0.12788 (0.04004)

All models adjust for the original eight controls. In the first three rows, income and education are fitted in separate regressions; the last row fits them jointly. Exclusions are diagnostics for unresolved coding, not a declaration that excluded records are incorrect. No significance stars are used to rank alternative samples.

Omitting the zero log income records reduces the income coefficient to -0.02334 , increases its standard error to 0.04651 , and gives a 95 percent interval from -0.11462 to 0.06793 . This does not establish that the true income association is zero. It shows that the full sample conclusion depends on how these low or specially coded values are handled. Their meaning needs to be recovered before the main income finding can be treated as stable.

The education estimate remains negative when the zero log income records are omitted (-0.13605 , standard error 0.04031), and when education code zero is omitted (-0.12691 , standard error 0.04329). That consistency is useful, but it does not validate the category spacing assumption or identify a causal education effect. The joint model addresses mutual adjustment for the two measured socioeconomic indicators without resolving unobserved differences between households.

7 3 Outcome construction diagnostic

The inverse transformation identity described in Section 4 is satisfied by 905 records and not by 14. All discrepancies make the exponentiated total smaller than the quantity implied by the subscales; the discrepant totals occur at boundary values. The original script contains a winsorization command, which may contribute to boundary differences, but a complete pipeline is unavailable. The pattern therefore does not prove that the questionnaire responses are invalid or that winsorization explains every discrepancy.

As a diagnostic only, fitting the adjusted models in the 905 records that satisfy the identity gives an income coefficient of -0.05940 (standard error 0.01732) and an education coefficient of -0.09750 (0.03921). This subset is not a certified clean sample. Selecting records by an outcome consistency condition can itself change the distribution, and the assumed offset identity may not be the intended

construction rule. The appropriate next step is to recover the item scoring and transformation sequence, rather than silently delete observations.

8 Discussion

The revised analysis supports a limited conclusion: in the available Ya'an analytical extract, education and household income are negatively associated with the stored caregiver distress outcomes under the original adjustment set. The main income results are numerically reproducible. Reproduction, however, is not the same as demonstrating that the data construction is correct or that the inference is stable. The zero log income diagnostic materially weakens confidence in a simple, general income gradient in this sample.

Education shows greater consistency across the selected checks, although its coded scale remains imperfectly documented. Educational attainment is also not a readily randomized exposure. Differences in family background, employment, migration, health and social resources could be related to both schooling and current distress. The result should therefore be understood as a socioeconomic association requiring better measurement and identification, not evidence that education directly supplies emotional intelligence or effective coping.

The caregiver subgroup analysis illustrates the importance of restrained interpretation. Mothers form most of the analytical sample, while fathers are a small group. The tables can describe estimated associations within each group, but they do not demonstrate that mothers have worse mental health or that grandparents are more vulnerable to changes in income. Neither significance patterns nor intercept comparisons provide that evidence. A better comparison would define an estimand, use a pooled specification and report uncertainty about the difference.

For practice, the study supports paying attention to the resources and circumstances surrounding caregivers when designing future evaluations. It does not identify a particular income, education or parenting intervention as effective. A useful next study could measure financial strain, caregiving time, available support and service access directly, then examine how those conditions relate to changes in symptoms over time. Evaluating a specific support program would require a design that separates its effect from baseline differences.

The original ambition to connect socioeconomic circumstances, caregiver well being and child development remains valuable as a research direction. This dataset does not establish that chain. A future study would need independently measured child outcomes, temporally ordered caregiver measures and an appropriate approach to confounding before investigating mediation. Subjective social status could also add information, but its role cannot be inferred from household income or education alone.

8 1 Limitations

First, the data are cross sectional and geographically restricted. Neither the recruitment process nor the complete exclusion flow is reconstructed from the analytical worksheet. Complete cases can differ systematically from the recruited population, and the local village sample does not establish

representativeness for rural China. No causal effect of socioeconomic status or the wider intervention is identified.

Second, several construction choices remain unresolved. The raw questionnaire items, total score assembly, offsets and full winsorization sequence are not recovered. Income values at zero on the stored log scale and education code zero need clarification. The revision reports stored variables and makes these uncertainties visible instead of asserting raw scale meanings or substituting guessed numbers. Until the source coding is documented, the results should be read as associations in an archived analytical extract.

Third, the standard errors address heteroskedasticity but not village dependence. This affects confidence intervals and significance statements, especially for subgroup models. The extract also lacks verified sampling weights and a complete design record. The numerical checks validate the specified calculations, not the survey design, theoretical explanation or adequacy of the standard error model.

Fourth, the models use a limited set of controls, and some numerical category codes are treated linearly. Self reported health may overlap conceptually with distress, while expected child education may reflect circumstances related to the outcomes. Alternative specifications should be guided by a clearly defined question and a documented measurement model. Adding controls indiscriminately would not itself solve confounding.

Finally, component, subgroup and audit analyses involve multiple related comparisons and were not preregistered. The additional coding checks were selected during this revision. They identify vulnerabilities and priorities for further work; they do not supply a comprehensive robustness guarantee. No independent statistical or clinical reviewer has approved the revised analysis.

9 Conclusion

In the archived 919 record sample, higher household income and higher caregiver education codes accompany lower stored caregiver distress outcomes in the main cross sectional regressions. Education remains negatively associated with distress across the selected coding checks. The income estimate is substantially more uncertain after omitting 16 records at zero on its stored log scale, making the interpretation of those records a central unresolved issue.

The findings describe caregiver symptoms, not child outcomes, happiness or diagnosed disorders. They do not show that increasing income or education would cause mental health to improve, and they do not establish group differences between fathers, mothers and grandparents. A defensible next step is to recover the questionnaire and preprocessing documentation, resolve the special codes, and reassess inference using the original sampling structure. This would provide a firmer basis for studying how household resources and caregiving conditions relate to well being.

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Appendix A Variable definitions and unresolved construction

The definitions below combine the original thesis, its recovered variable notes and the archived worksheet. They distinguish an observed analytical code from a verified raw measure. No individual survey records, respondent identifiers or private file locations are included in this manuscript.

Table A1 Recovered analytical variable definitions

Variable	Meaning in this revision	Documentation limit
dass_total	Stored log total distress outcome	Raw item summation, offset and any capping require reconstruction. Fourteen records fail the candidate inverse score identity.
dass_stress dass_anxiety dass_depression	Stored log component outcomes	Original thesis identifies DASS 21. Exact raw scoring and multiplier are not certified from this extract.
income	Stored log household income	Sixteen values equal zero. Original amount, unit, offset and special coding must be confirmed; zero is not assumed to mean missing or no income.
edu	Ordinal caregiver schooling code 0–5	Recovered legend: 1 no schooling; 2 primary; 3 junior middle; 4 high school level; 5 college or above. Code 0 is unresolved and occurs 23 times. Codes are not years.
inter	Internet access indicator	Stored 0/1 variable; detailed questionnaire wording not recovered.
air	Air conditioning access indicator	Stored 0/1 variable.
doc	Village doctor indicator	Stored 0/1 variable; availability and service quality are not interchangeable.
car	Household car ownership indicator	Stored 0/1 variable.
age	Stored log caregiver age	Original notes identify a logarithm. Raw age construction and any winsorization require source confirmation.
health	Self reported caregiver health indicator	Stored 0/1 variable; original narrative interprets 1 as better health, but item wording should be checked.
csex	Child sex indicator	Stored 0/1 code; does not identify the caregiver's sex. Detailed value labels should be confirmed.
cstudy	Child education expectations code	Recovered legend: 1 moderately anticipated; 2 anticipated; 3 highly anticipated. The exact questionnaire wording and scale construction still need confirmation.
who	Primary caregiver relationship	Recovered notes: 1 father; 2 mother; 3 grandparent. Sample counts 89, 654 and 176.
tc	Archived age restriction flag	Stored 0/1 code. The tc = 0 restriction retains 907 records. Detailed creation should be checked before describing it as a verified threshold rule.

Priority checks are to recover the questionnaire, value labels and complete preparation script; confirm the income boundary values and undefined education code; reconstruct the DASS component and total scores; reconcile the original 922 and 919 samples; and restore village identifiers and the sampling design. These are unresolved research tasks, not completed checks.

Appendix B Numerical verification and revision record

Nine income specifications were compared with the final 2024 PDF: the unadjusted and adjusted total outcome models, three adjusted component models, three caregiver subgroup models and the recorded age restriction model. In every case, the focal coefficient, observation count and R squared agree with the precision printed in the original table. The adjusted income coefficient is -0.06353189 before rounding. The recovered extract therefore supports these calculations, but does not independently validate how the extract was created.

The revised education models are newly computed on the common 919 record extract. The adjusted total coefficient is -0.13209086 , compared with -0.12746 in the original 922 record table. All revised regressions use HC1 standard errors, including the unadjusted models. Coefficients and covariance calculations were checked using both a statistical library and separate matrix calculations; this is a numerical consistency check, not independent expert review.

The original fixed effects notation was replaced by cross sectional OLS equations. Labels, sample counts and interpretations were aligned with the available source. Unsupported causal, clinical, happiness, child outcome and between group claims were removed or narrowed. The impossible raw income standard deviation was not repaired by guessing. Bibliography entries not used in the revised argument were omitted, and cited entries were checked against primary publisher or bibliographic records.

The editable archive also contains unfinished entropy weighting and discussion material that is absent from the final 25 page PDF. This revision follows the scope of the final PDF rather than inserting those abandoned draft sections. It adds explicitly labeled coding diagnostics because the recovered data raise new questions about the original conclusions. The original thesis remains a separate historical record.

AI assistance was used in reviewing inconsistencies, checking source records, preparing analysis code and drafting this revision. The revised manuscript remains for the author's review. It has not been approved by the original university, data custodians, a statistician or a clinical reviewer. Raw respondent data are not included, and no new ethics approval or permission to redistribute participant data is claimed.